

Ulises Javier Alvarenga

Unity Game Developer · VR · Multiplayer · AI Systems · Open to Remote

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EXPERIENCE

VR / XR Game Developer · [Studio Soup](#)

Jan 2024 – Present

- ▶ **Engineered a real-time LLM-powered NPC pipeline in VR:** voice input → speech-to-text → constrained LLM (DeepSeek / lightweight GPT) → TTS output. Designed prompt constraints to enforce domain scope and token limits on free-tier APIs; implemented manual context injection per request (stateless API), with error catching, retry logic, and graceful fallback responses to meet real-time VR latency requirements.
- ▶ **Drove VR performance to a stable 72 FPS on Meta Quest standalone:** diagnosed GPU-bound bottlenecks (GPU ~99%, CPU ~60%) using Unity Profiler, OVR Metrics Tool, and Frame Debugger; resolved through draw call reduction (static/dynamic batching, LOD Groups, dynamic occlusion culling), texture compression, material downgrading (Lit → Simple Lit → Unlit), and targeted shader optimization.
- ▶ **Architected and shipped Astropark (Meta Quest) end-to-end:** owned gameplay systems, VR interaction design, and performance pipeline using Unity + Meta SDK.
- ▶ **Built LAN multiplayer VR sessions via NGO:** implemented server-authoritative state synchronization for player positions, interactions, and gameplay events across networked clients.
- ▶ Delivered Wonderland on Meta Horizon and shipped AR experiences via Snapchat Lens Studio, adapting systems to each platform's spatial tracking and runtime constraints.

Unity Developer · [Freelance](#)

Jun 2021 – Present

- ▶ **Designed and shipped a server-authoritative multiplayer racing demo using NGO + Steam:** implemented position sync, interaction events, and chose server authority deliberately to eliminate client-side inconsistencies in a semi-competitive co-op context.
- ▶ Refactored and stabilized production Unity codebases (2D, 3D, VR) across multiple client projects — reducing regression risk, improving runtime stability, and lowering onboarding overhead through cleaner architecture.
- ▶ Delivered CPU/GPU profiling passes on mobile and VR targets, identifying and resolving batching, overdraw, and asset pipeline inefficiencies.

Unity Developer · [F.a.R Team](#)

Nov 2022 – Jan 2024

- ▶ **Led codebase refactor of Room Makers:** applied SOLID principles and GoF design patterns (Factory, Observer, Singleton, Command) to decouple systems, increase reuse, and reduce regression surface. Collaborated via Agile workflows (Jira, Trello, Git).

SKILLS

Engine & Platforms Unity (VR/XR, NGO, procedural systems, custom editor tools, REST API, UI, physics) · Meta SDK · Meta Horizon · Lens Studio

AI / LLM Integration Voice-to-LLM-to-TTS pipeline design · Prompt engineering · Context management · Fallback & error handling · Latency-aware API design

Performance GPU/CPU profiling (Unity Profiler, OVR Metrics, Frame Debugger) · Draw call optimization · LOD · Occlusion culling · Texture compression · Shader tuning

Multiplayer NGO (Netcode for GameObjects) · Server-authoritative architecture · State sync · Steam integration

Languages C# (primary) · Python · TypeScript

Architecture SOLID · GoF design patterns · Component-driven / modular systems design

Tooling Git · Jira · Trello · Agile / Scrum

EDUCATION & TRAINING

B.Sc. Computer Science · National University of Quilmes

2022 – Present

Design Patterns for Videogames · Daniel Parra

Feb – Mar 2022

SOLID principles, GoF patterns, applied Unity architecture.

LANGUAGES

Spanish — Native · **English** — B2 Certified, Professional Proficiency

PROJECTS & TOOLS

Vlys Scene Notes Lite · [Unity Asset Store](#) · [Editor Tooling](#)

Released

- ▶ Designed and published a Unity editor tool for in-scene annotation, completing the full release pipeline: development, API integration, packaging, documentation, and Asset Store submission.

Unity-Agent-Expert · [AI](#) · [Multi-Agent Systems](#)

github.com/Ulysses-Alv/Unity-Agent-Expert

- ▶ **Built a multi-agent orchestration system to improve reliability of LLM-generated Unity code:** domain-specific agents grounded in parsed Unity 6000.3 LTS documentation feed into an orchestrator layer, reducing outdated or incorrect code outputs from general-purpose models.
- ▶ Addresses a concrete production problem — LLMs hallucinating deprecated Unity APIs — through structured context injection and agent specialization rather than fine-tuning.

Shader Graph MCP Tool · [AI](#) · [Unity Tooling](#)

In Development · *github.com/Ulysses-Alv/unity-ai-shader-graph*

- ▶ Extending Unity Shader Graph workflows via MCP-based integrations; exploring AI-assisted shader authoring as a practical developer tooling problem.

Papanotifier · [Systems Integration](#) · [Private](#)

- ▶ **Architected a cross-platform real-time notification system:** Telegram bot frontend, Firebase backend for message persistence and routing, and Smart TV delivery — integrated as a unified event-driven pipeline.

BocaQueue · [Automation](#) · [Concurrency](#) · [Private](#)

- ▶ **Built a multi-session automation system using headless environments:** manages concurrent session lifecycle, exposes a screenshot-based selection interface for session state inspection, and handles parallel workflow execution with process isolation.